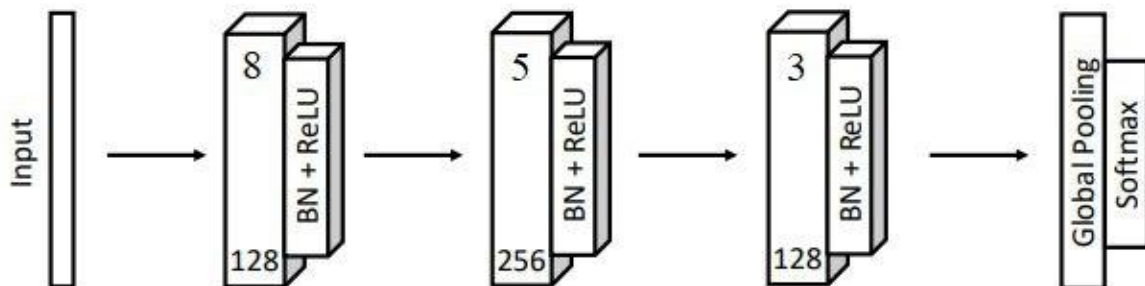


Resultados experimentos com FCN

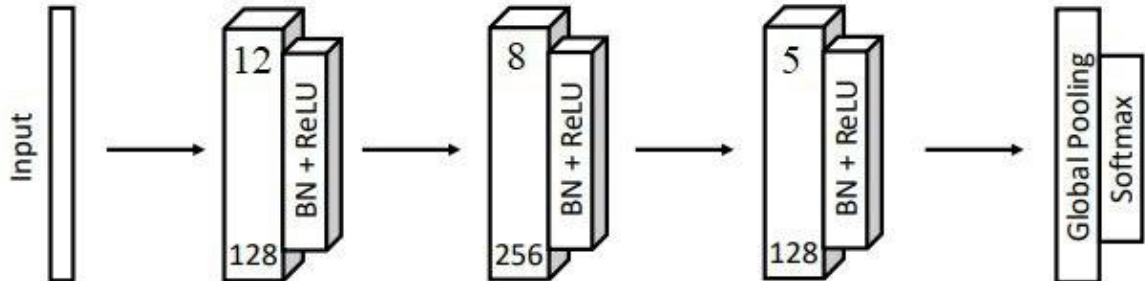
1) Modelos:

Quando conversamos pelo Slack, você tinha me orientado para testar um FCN com kernels maiores, por causa disso, te avisei que iria testar um FCN com kernels [12,8,5] em vez de [8,5,3], como é no padrão. No entanto, acabei me empolgando com o PyTorch e testei, além dessa mudança, outras 2.

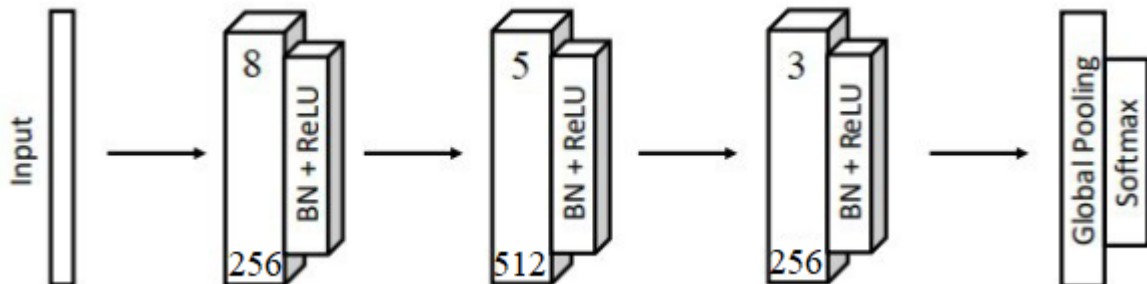
- FCN (padrão):



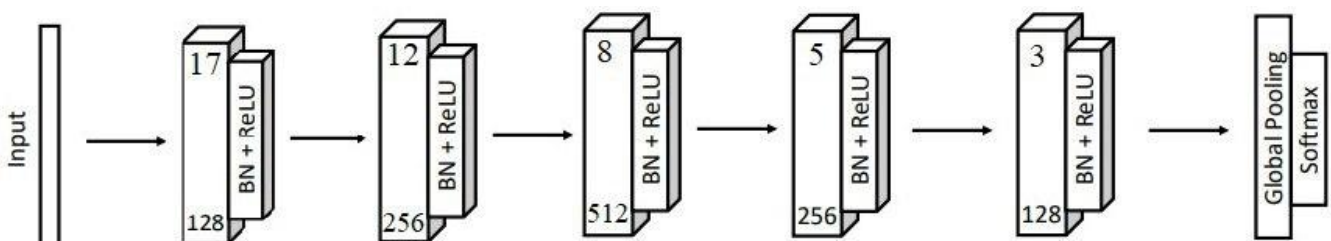
- FCN_biggerKernels:



- FCN_moreKernels:



- FCN_moreLayers:



2) Minha interpretação:

Na maioria dos casos, o FCN padrão se dá melhor ou empata que qualquer uma das modificações, seja em classificação ou regressão, dataset univariado ou multivariado.

Nos casos em que as modificações do FCN empatam com a versão padrão, eles o fazem num tempo maior.

Estou mandando todos esses prints, caso você queira dar uma olhada.

3) Resultados:

Fiz experimentos tanto para classificação quanto para regressão extrínseca. Para ambos, selecionei 2 datasets univariados e 2 multivariados, e rodei o mesmo modelo 5 vezes para cada um dos dados usando Early Stopping.

Para classificação, usei os datasets ArrowHead e OSULeaf como univariados e NATOPS e Epilepsy como multivariados.

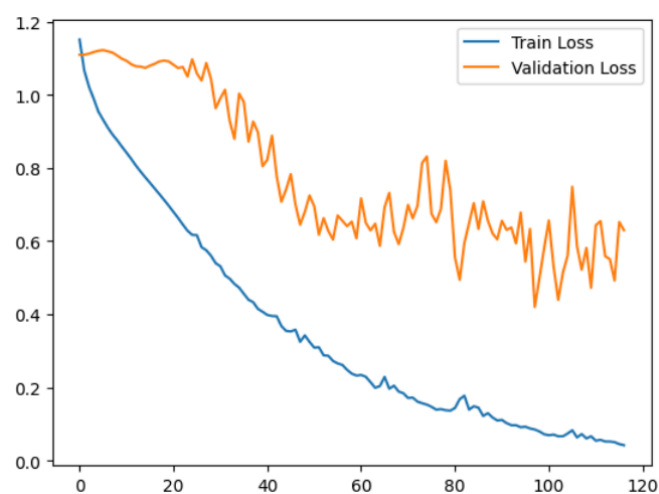
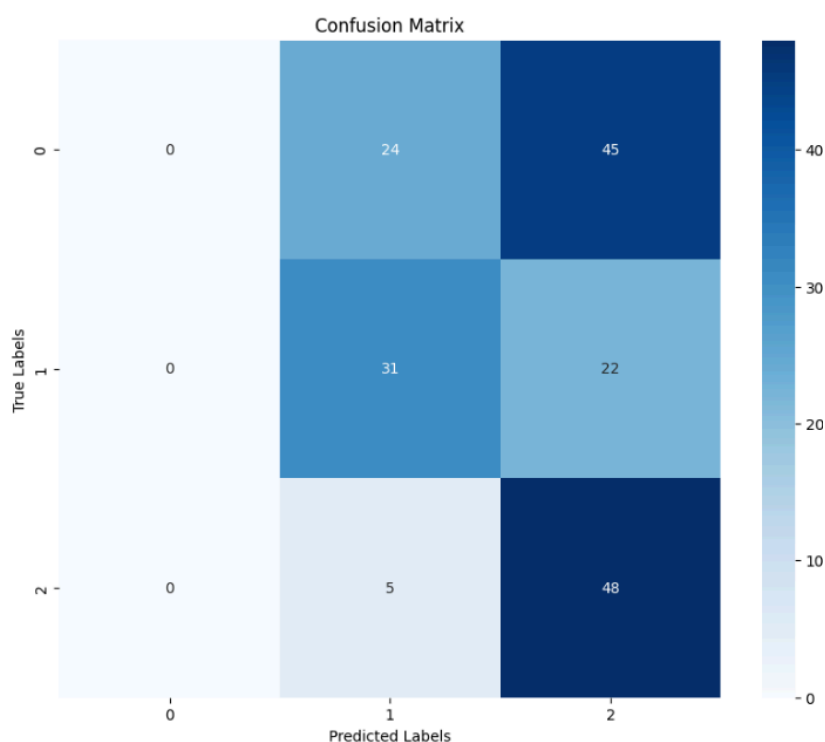
Por outro lado, selecionei para a regressão extrínseca FloodModelling1 e Covid2Month como univariados e AppliancesEnergy com HouseholdPowerConsumption2 como multivariados.

Abaixo estou apresentando a média e o desvio padrão dos resultados das 5 vezes que rodei cada modelo. A matriz de confusão e o gráfico das losses estão todas muito parecidas entre elas.

a) Classificação:

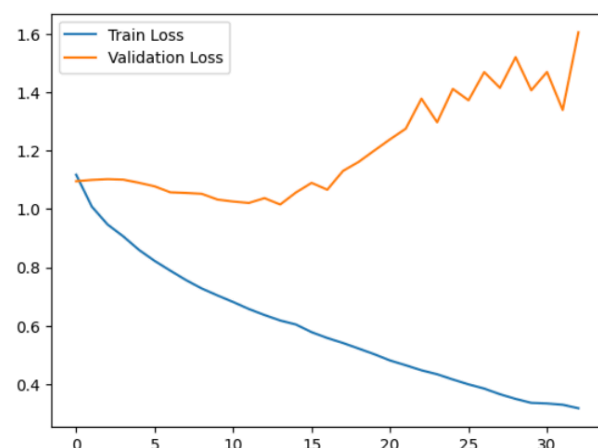
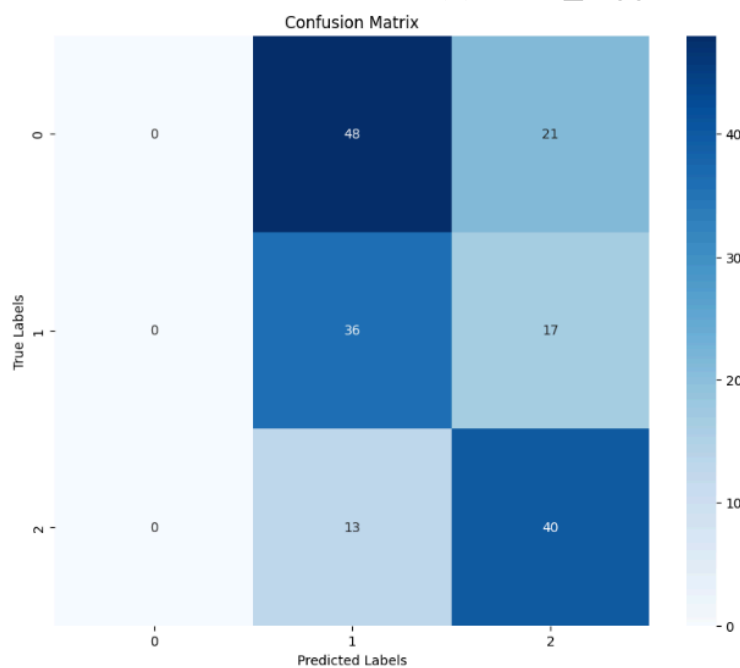
i) ArrowHead (univariado):

(1) FCN:



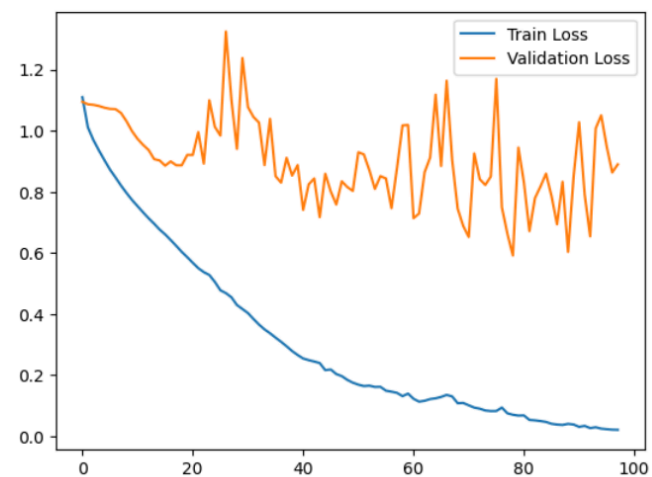
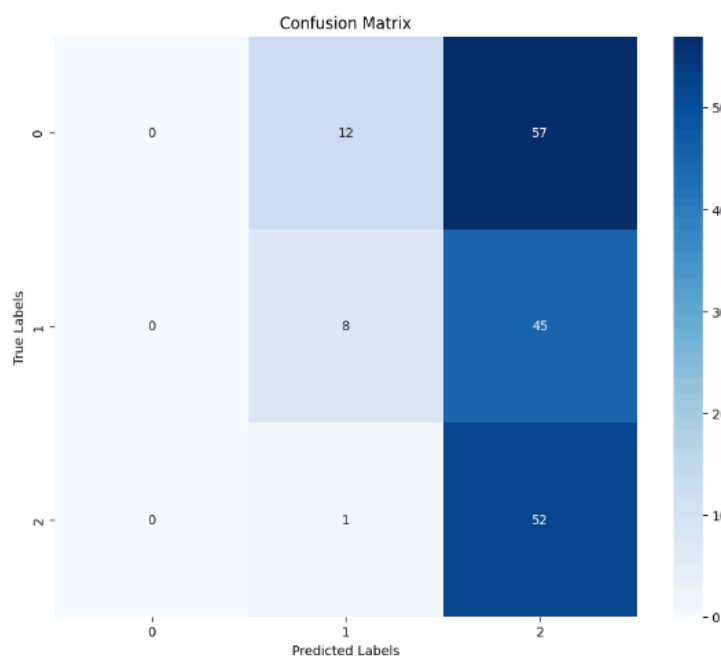
Acurácia: 0.56 +- 0.15
F1 Macro: 0.53 +- 0.17
F1 Micro: 0.56 +- 0.15
Time (sec): 1.57 +- 1.63

(2) FCN_biggerKernels:



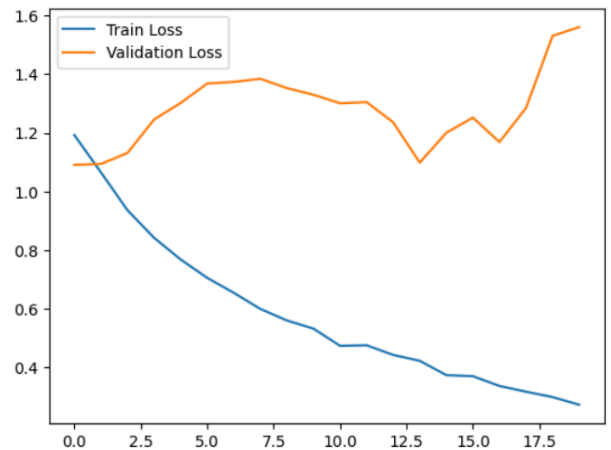
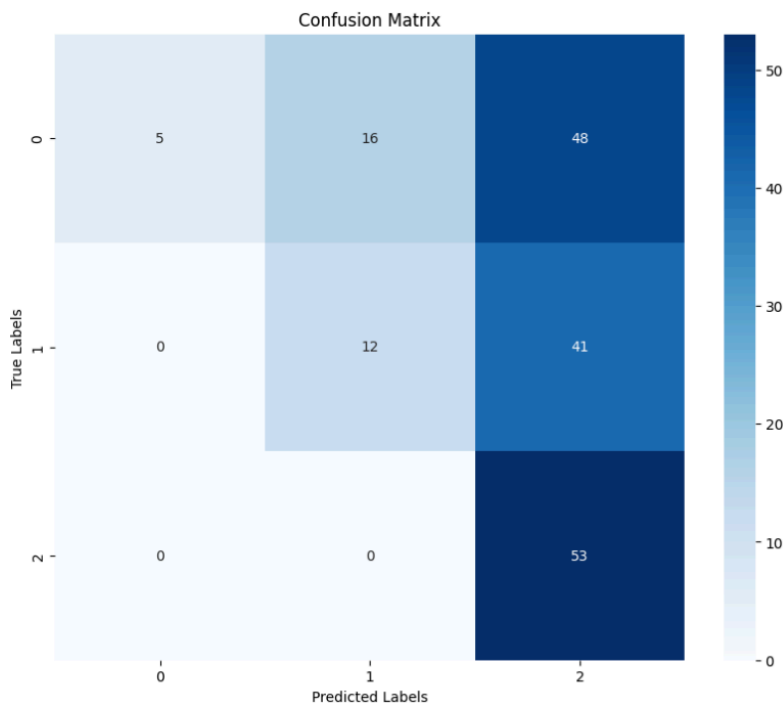
```
Acurácia: 0.41 +- 0.06
F1 Macro: 0.32 +- 0.09
F1 Micro: 0.41 +- 0.06
Time (sec): 0.63 +- 0.34
```

(3) FCN_moreKernels:



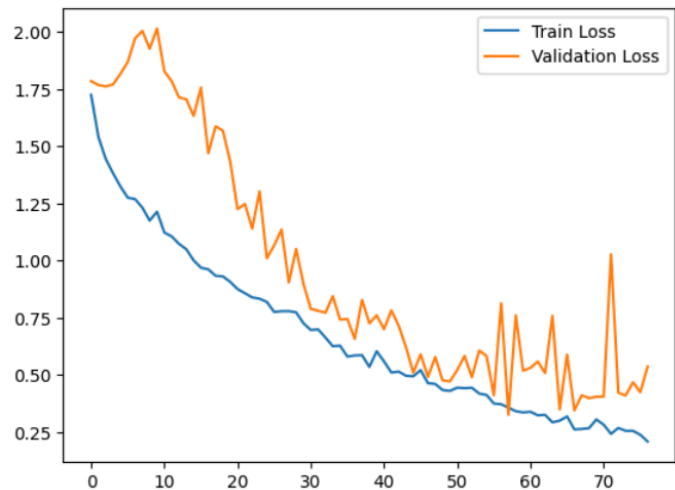
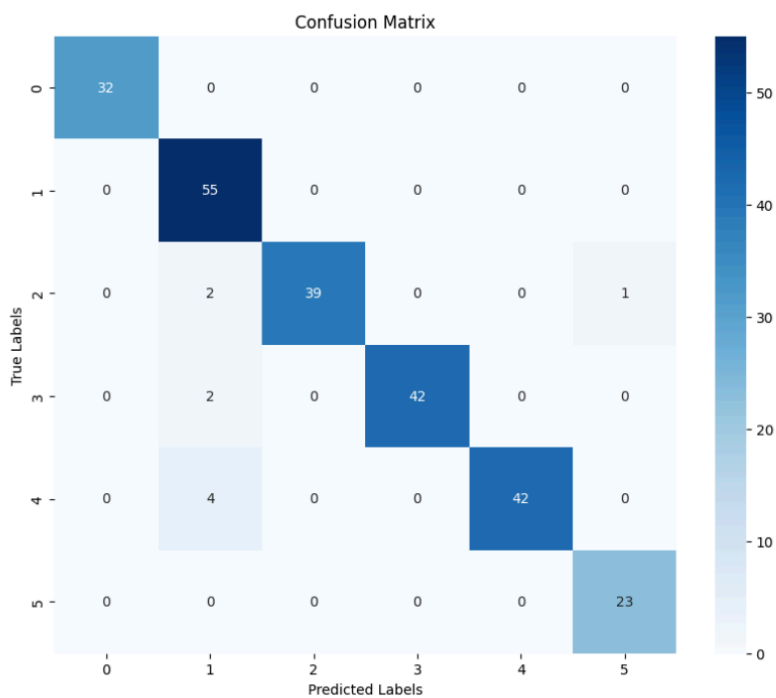
```
Acurácia: 0.39 +- 0.16
F1 Macro: 0.29 +- 0.22
F1 Micro: 0.39 +- 0.16
Time (sec): 1.15 +- 0.51
```

(4) FCN_moreLayers:



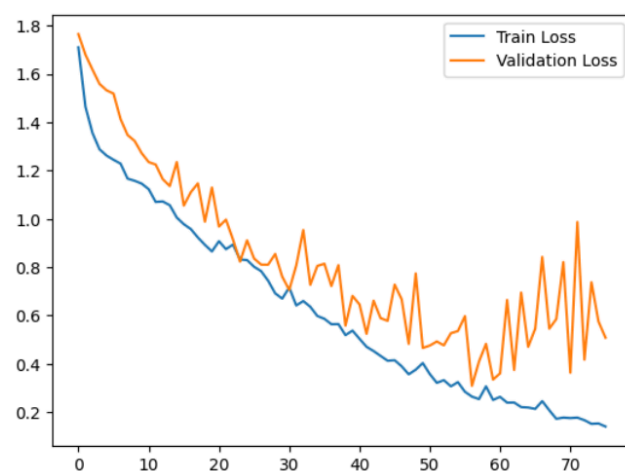
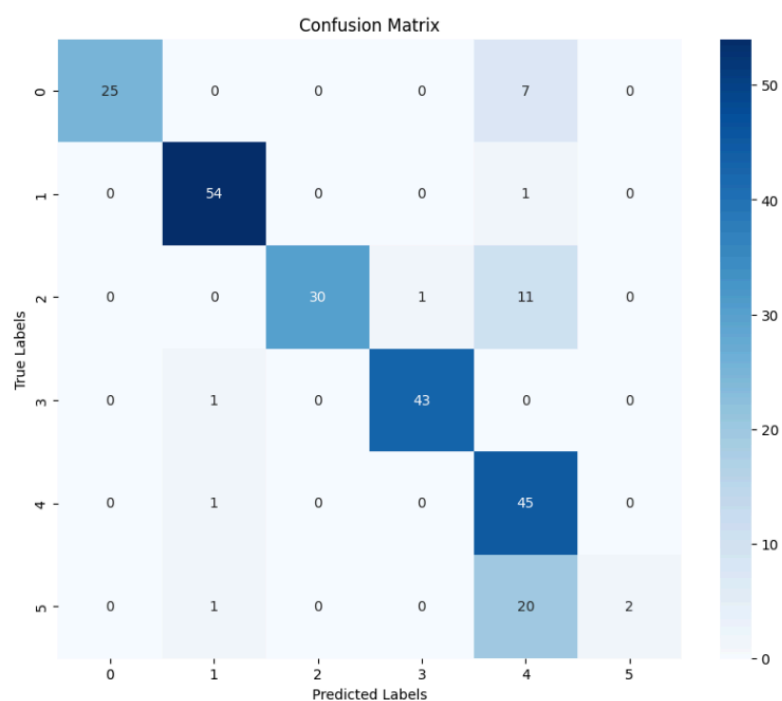
Acurácia: 0.37 +- 0.09
 F1 Macro: 0.27 +- 0.14
 F1 Micro: 0.37 +- 0.09
 Time (sec): 1.71 +- 1.41

ii) OSULeaf (univariado): (1) FCN:



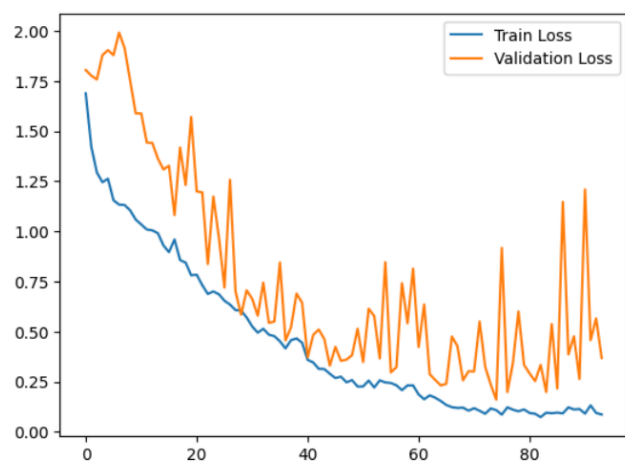
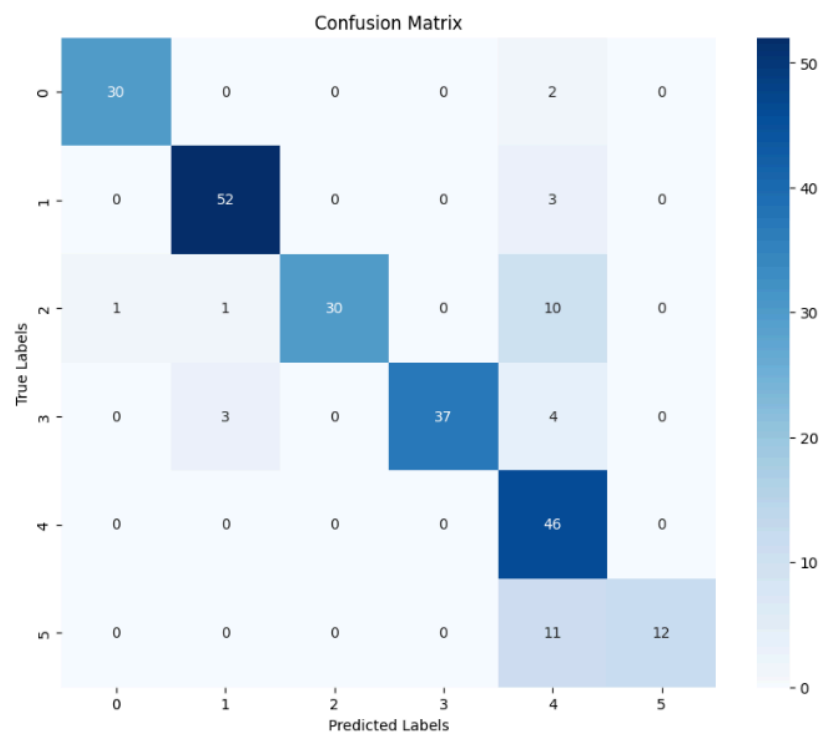
Acurácia: 0.87 +- 0.11
 F1 Macro: 0.87 +- 0.11
 F1 Micro: 0.87 +- 0.11
 Time (sec): 7.75 +- 2.72

(2) FCN_biggerKernels:



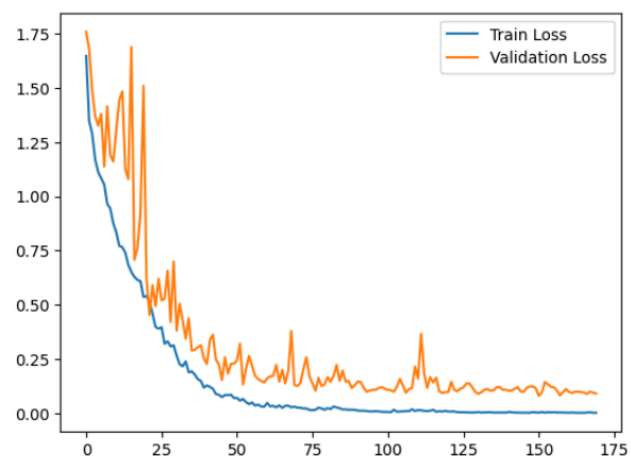
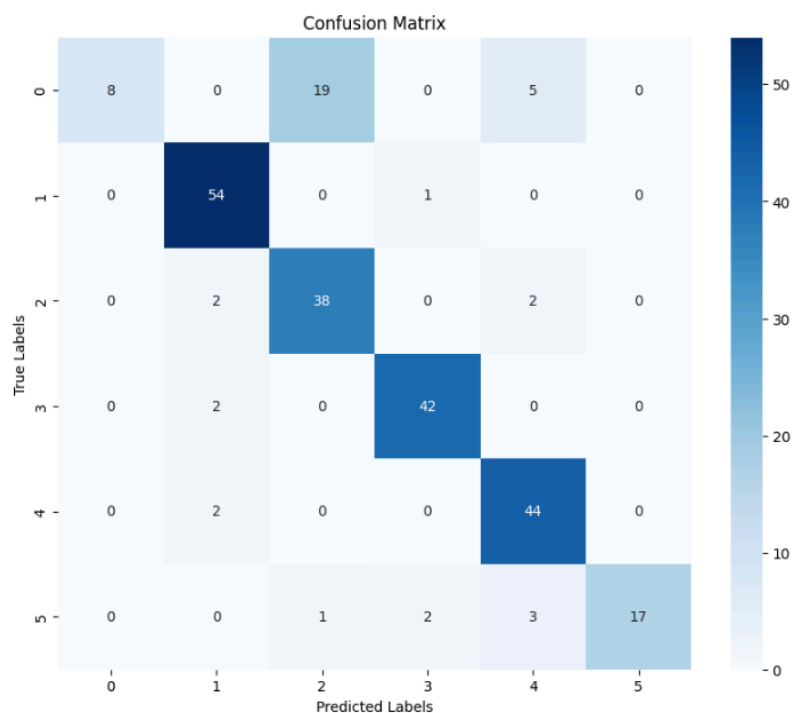
Acurácia: 0.82 +- 0.07
 F1 Macro: 0.79 +- 0.08
 F1 Micro: 0.82 +- 0.07
 Time (sec): 8.31 +- 1.58

(3) FCN_moreKernels:



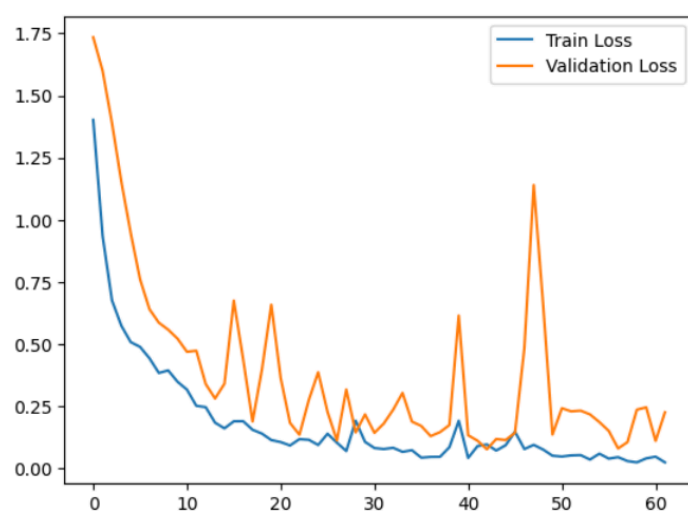
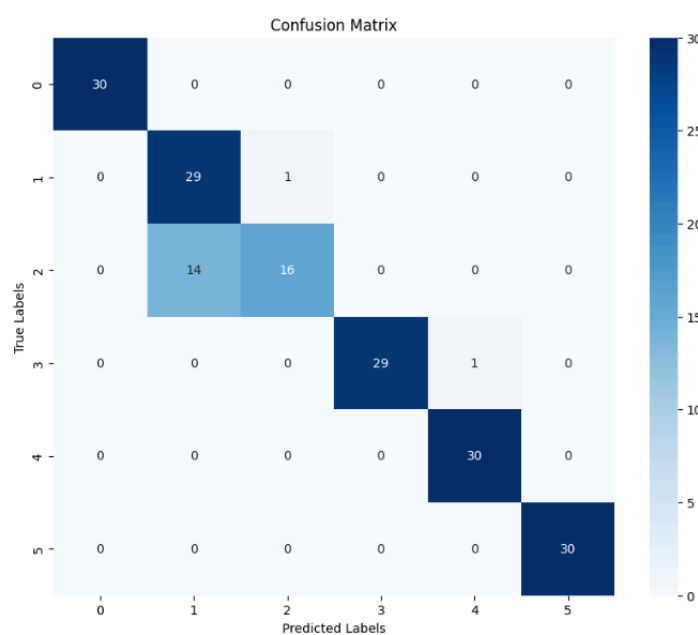
Acurácia: 0.85 +- 0.07
 F1 Macro: 0.86 +- 0.07
 F1 Micro: 0.85 +- 0.07
 Time (sec): 15.46 +- 3.96

(4) FCN_moreLayers:



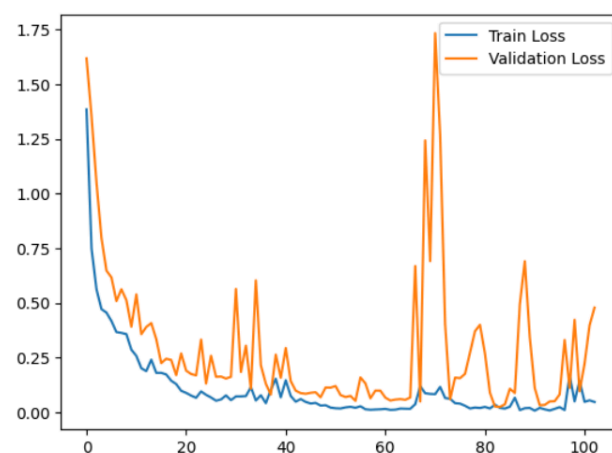
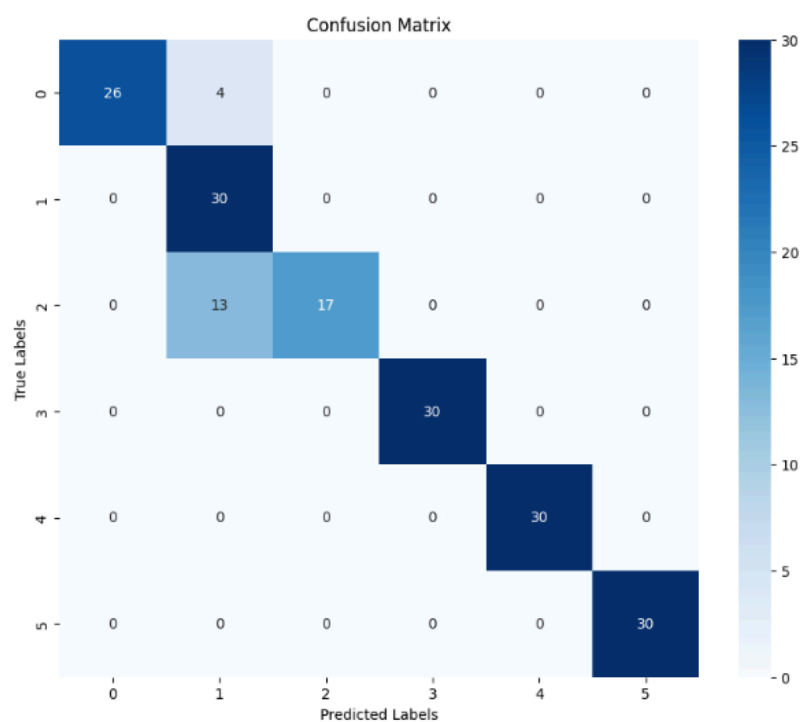
Acurácia: 0.9 +- 0.08
 F1 Macro: 0.89 +- 0.09
 F1 Micro: 0.9 +- 0.08
 Time (sec): 27.72 +- 8.23

iii) NATOPS (multivariado): (1) FCN:



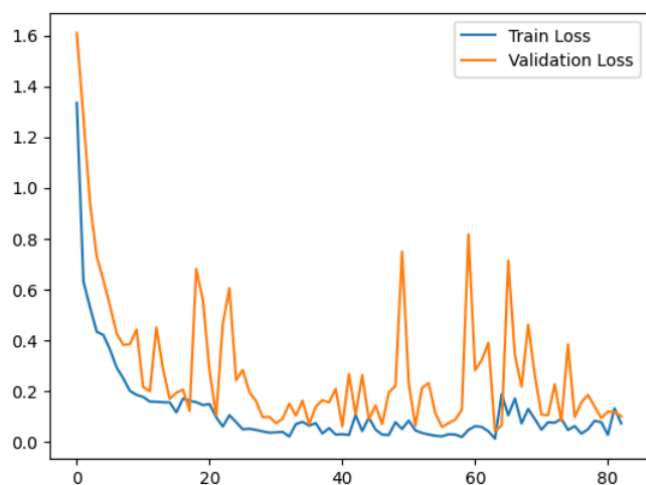
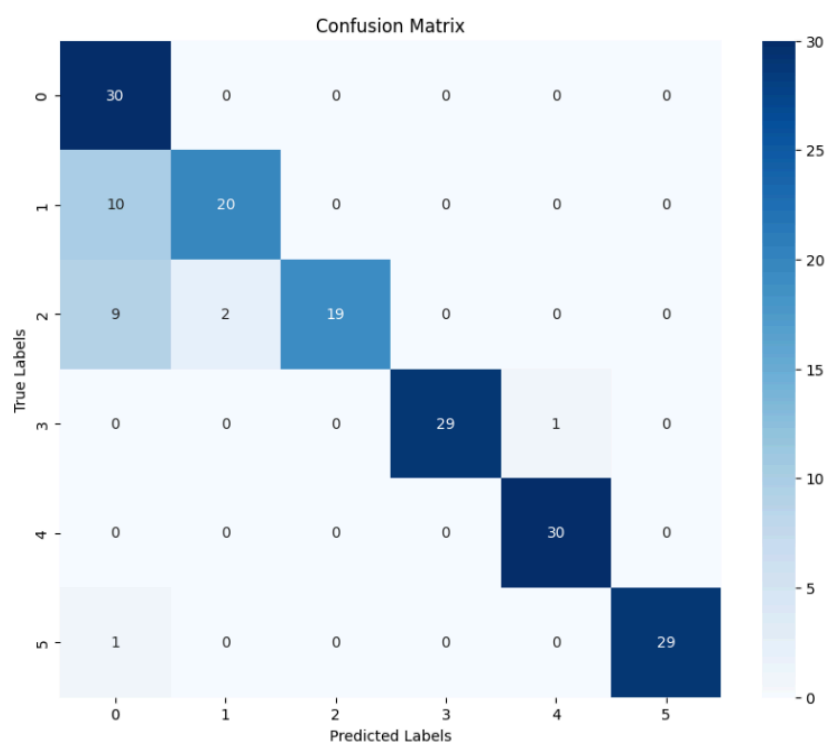
Acurácia: 0.91 +- 0.02
 F1 Macro: 0.91 +- 0.02
 F1 Micro: 0.91 +- 0.02
 Time (sec): 1.28 +- 0.12

(2) FCN_biggerKernels:



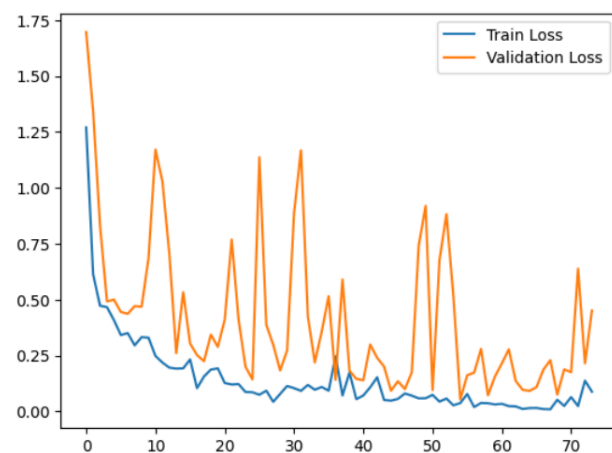
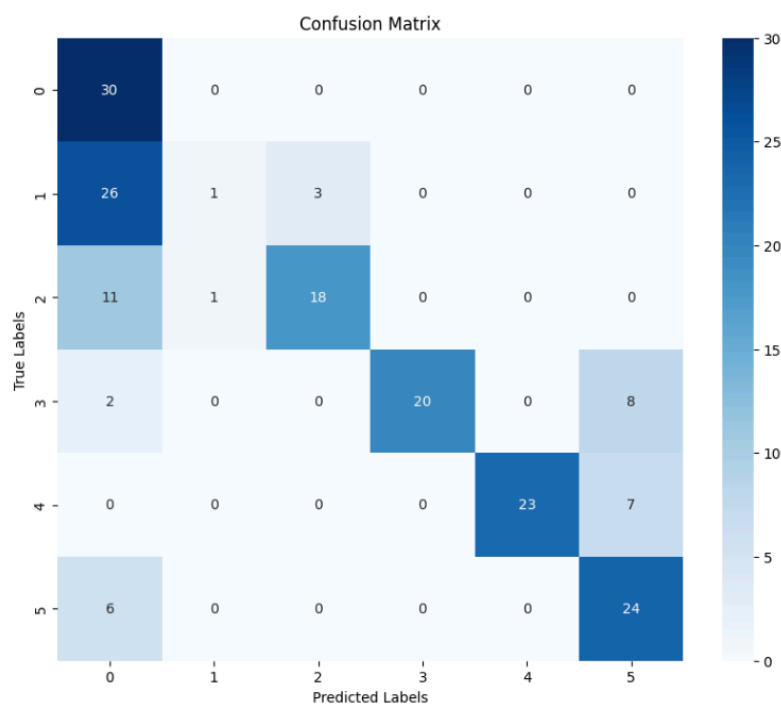
Acurácia: 0.9 +- 0.03
 F1 Macro: 0.89 +- 0.04
 F1 Micro: 0.9 +- 0.03
 Time (sec): 1.54 +- 0.23

(3) FCN_moreKernels:



Acurácia: 0.91 +- 0.02
 F1 Macro: 0.9 +- 0.02
 F1 Micro: 0.91 +- 0.02
 Time (sec): 2.01 +- 0.32

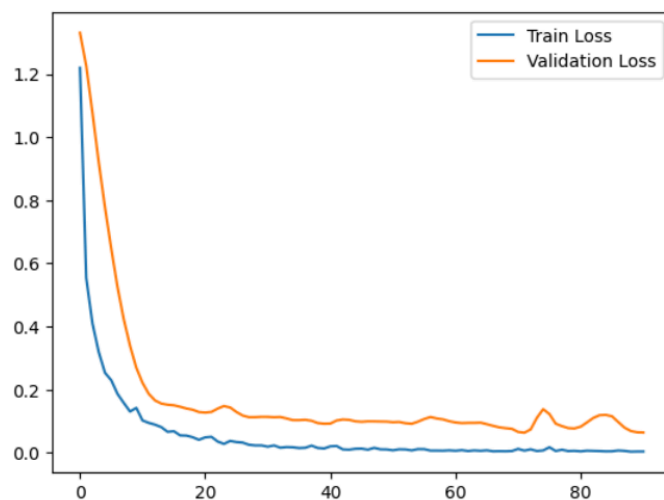
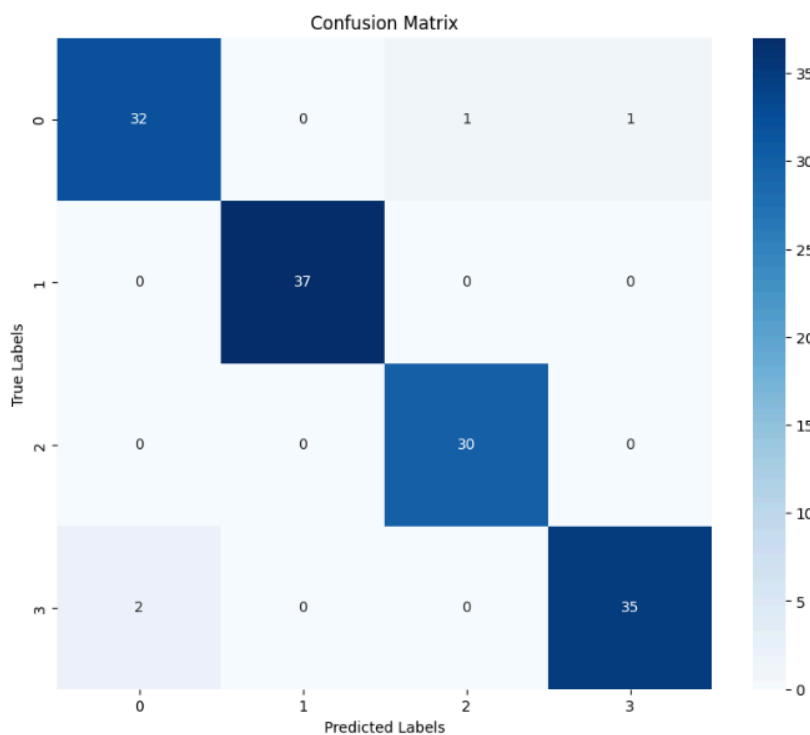
(4) FCN_moreLayers:



Acurácia: 0.85 +- 0.11
 F1 Macro: 0.83 +- 0.12
 F1 Micro: 0.85 +- 0.11
 Time (sec): 5.02 +- 2.33

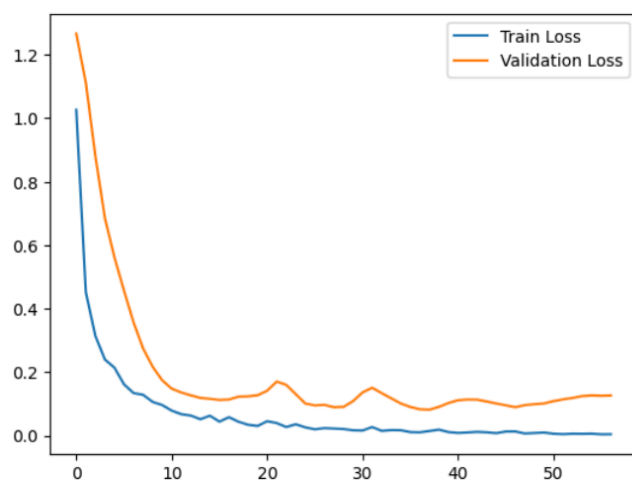
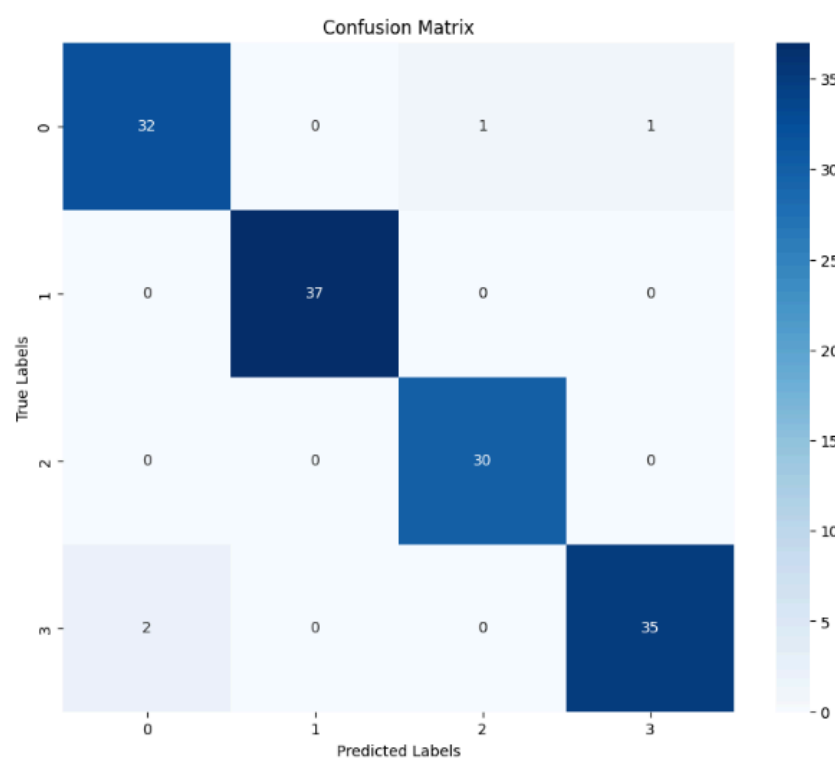
iv) Epilepsy (multivariado):

(1) FCN:



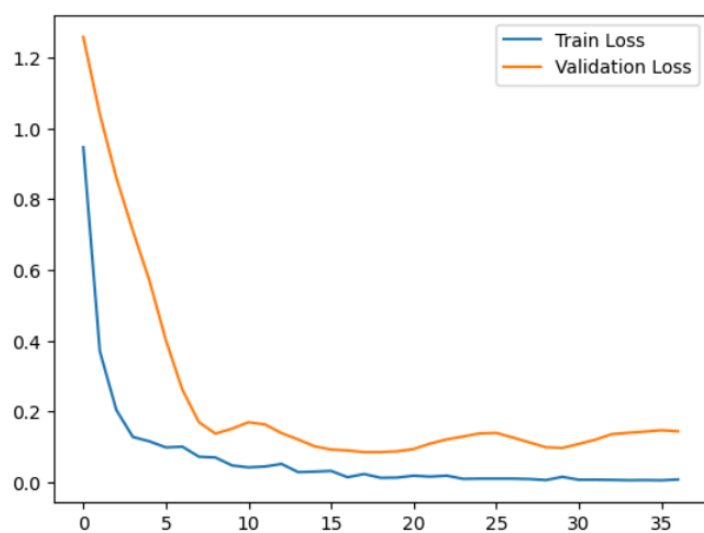
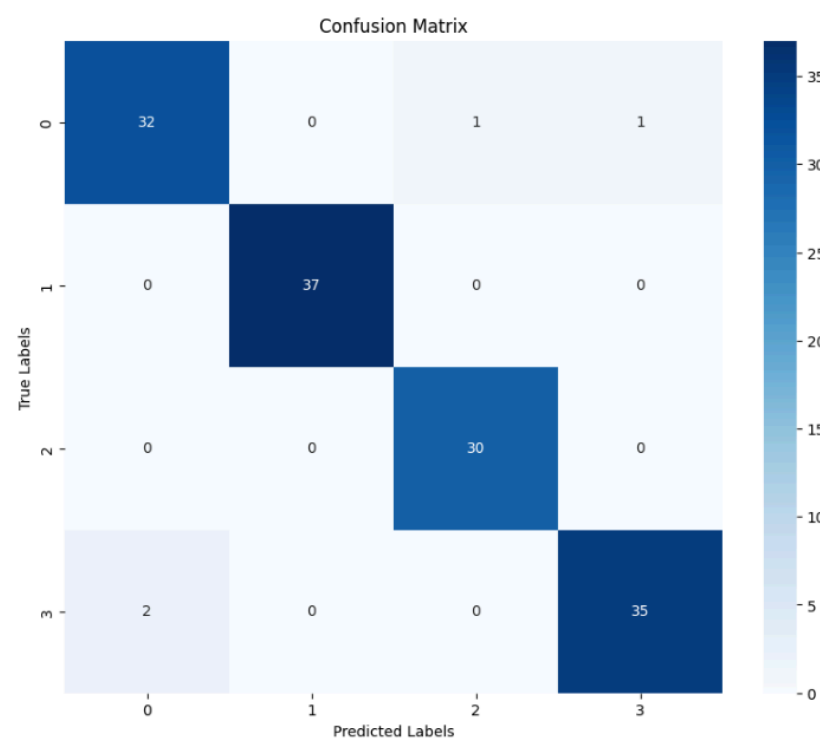
Acurácia: 0.97 +- 0.0
 F1 Macro: 0.97 +- 0.0
 F1 Micro: 0.97 +- 0.0
 Time (sec): 1.77 +- 0.31

(2) FCN_biggerKernels:



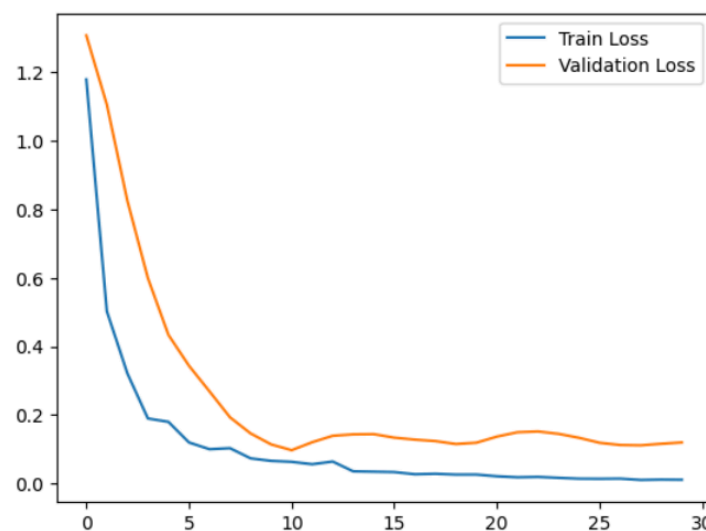
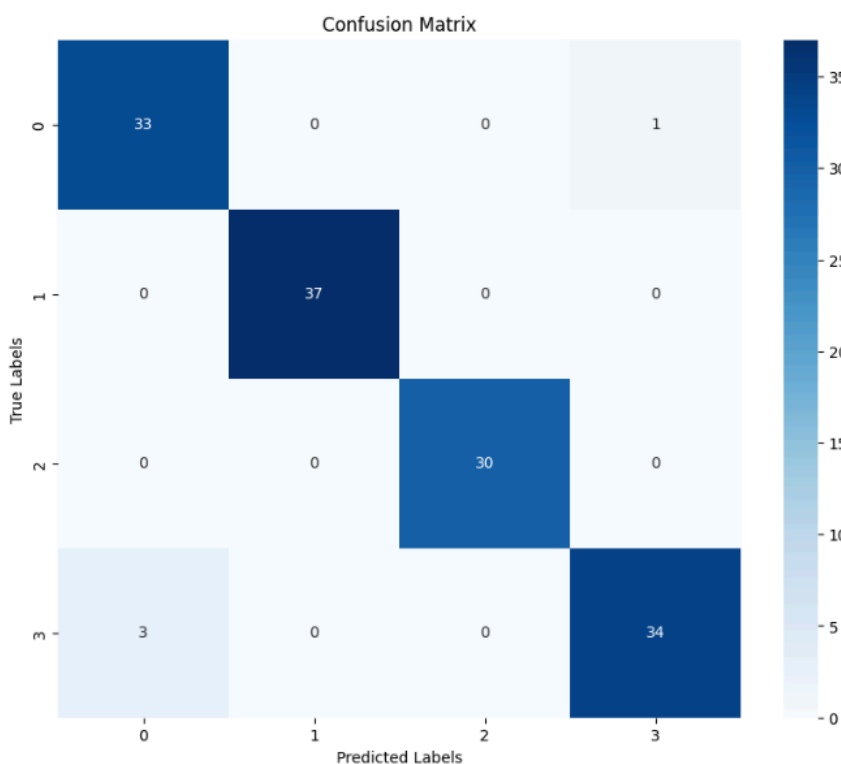
Acurácia: 0.97 +- 0.0
 F1 Macro: 0.97 +- 0.0
 F1 Micro: 0.97 +- 0.0
 Time (sec): 7.87 +- 3.7

(3) FCN_moreKernels:



Acurácia: 0.97 +- 0.0
 F1 Macro: 0.97 +- 0.0
 F1 Micro: 0.97 +- 0.0
 Time (sec): 3.39 +- 2.01

(4) FCN_moreLayers:



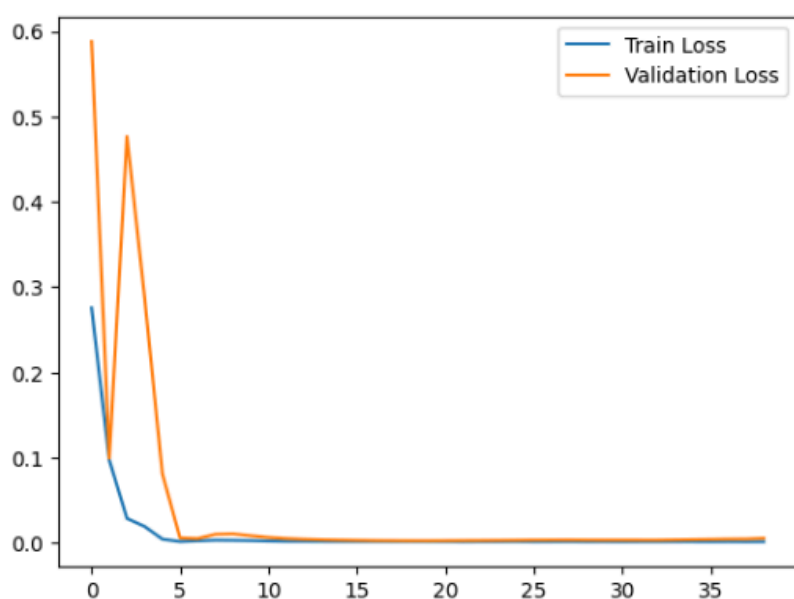
```

Acurácia: 0.97 +- 0.0
F1 Macro: 0.97 +- 0.0
F1 Micro: 0.97 +- 0.0
Time (sec): 7.14 +- 2.48
  
```

b) Regressão Extrínseca:

i) Covid3Month (univariado):

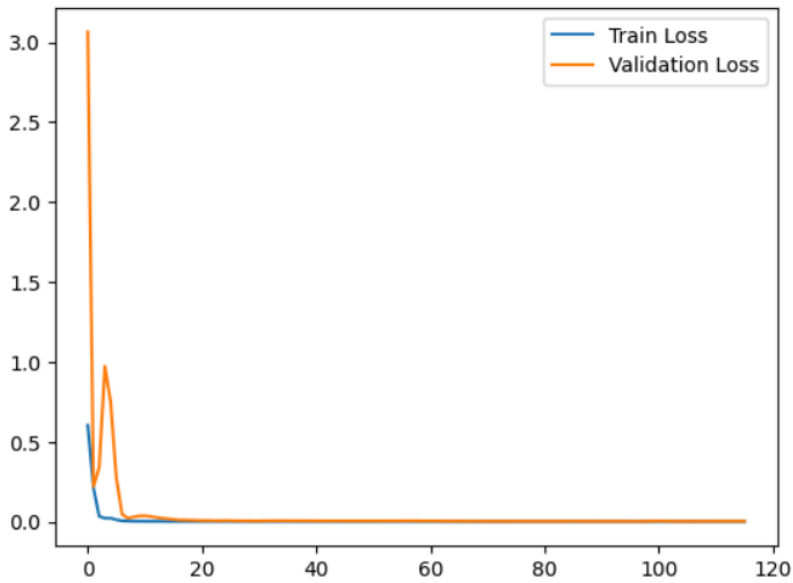
(1) FCN:



```

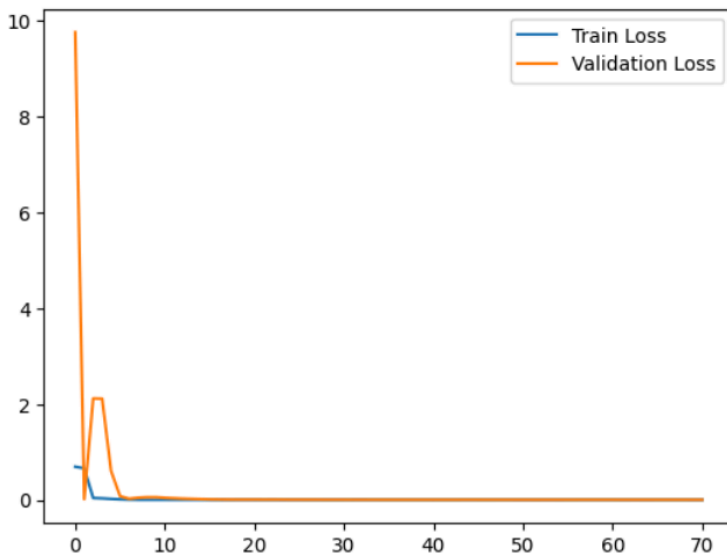
MSE: 0.0 +- 0.0
RMSE: 0.07 +- 0.01
MAE: 0.05 +- 0.01
Time (sec): 0.55 +- 0.05
  
```

(2) FCN_biggerKernels:



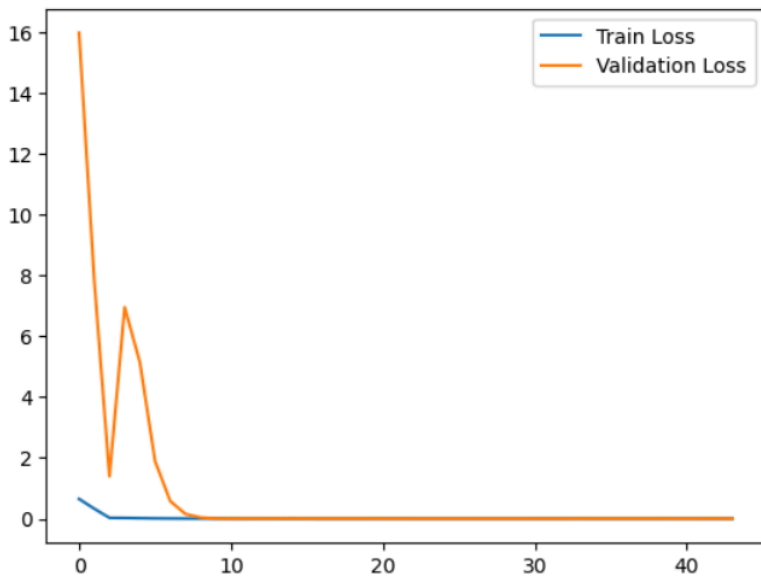
```
MSE: 0.0 +- 0.0
RMSE: 0.06 +- 0.02
MAE: 0.05 +- 0.02
Time (sec): 1.13 +- 0.61
```

(3) FCN_moreKernels:



```
MSE: 0.01 +- 0.0
RMSE: 0.07 +- 0.03
MAE: 0.06 +- 0.03
Time (sec): 2.6 +- 1.68
```

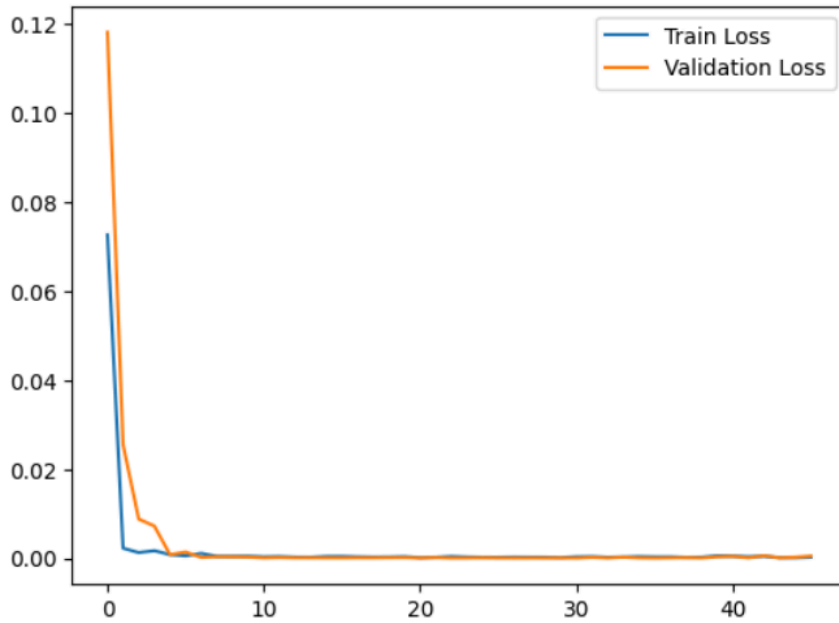
(4) FCN_moreLayers:



```
MSE: 0.0 +- 0.0
RMSE: 0.06 +- 0.01
MAE: 0.04 +- 0.01
Time (sec): 2.26 +- 0.32
```

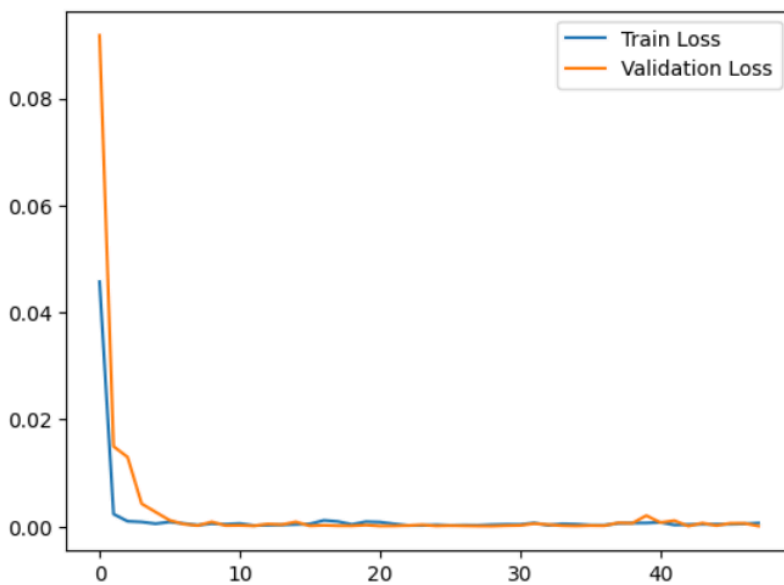
ii) FloodModeling1 (univariado):

(1) FCN:



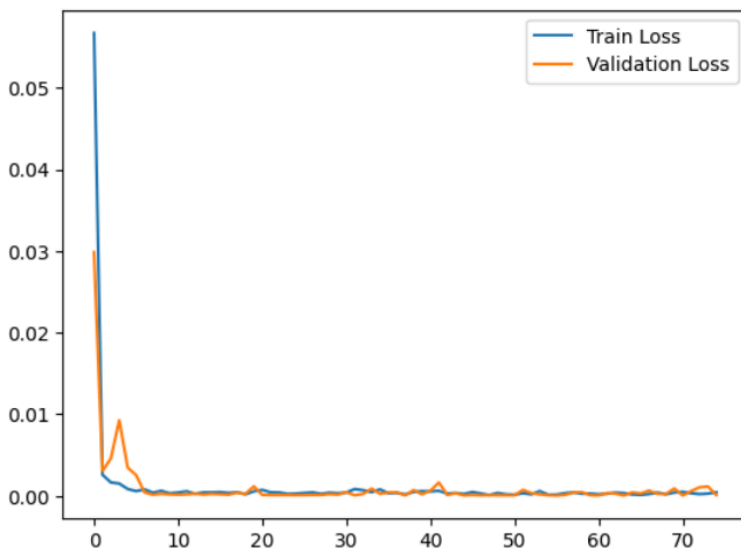
```
MSE: 0.0 +- 0.0
RMSE: 0.02 +- 0.01
MAE: 0.01 +- 0.01
Time (sec): 7.9 +- 3.11
```

(2) FCN_biggerKernels:



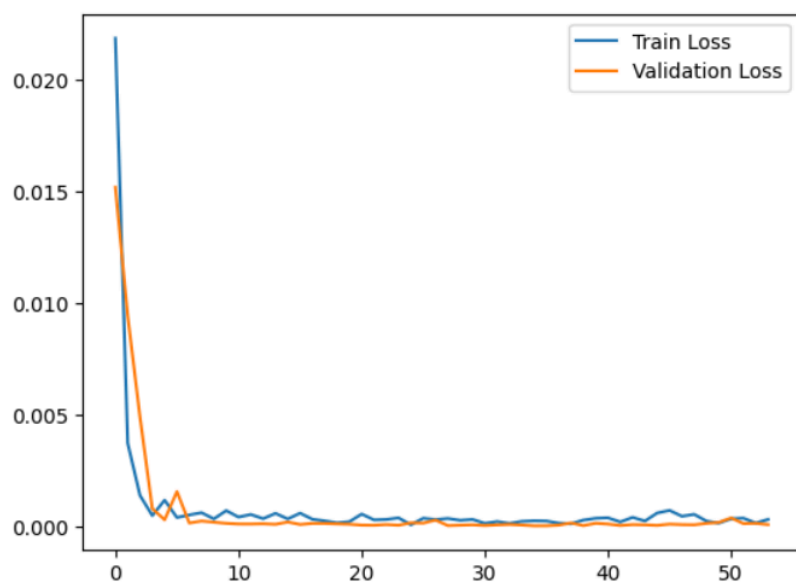
```
MSE: 0.0 +- 0.0
RMSE: 0.02 +- 0.01
MAE: 0.02 +- 0.01
Time (sec): 8.28 +- 1.92
```

(3) FCN_moreKernels:



```
MSE: 0.0 +- 0.0
RMSE: 0.02 +- 0.01
MAE: 0.02 +- 0.01
Time (sec): 16.81 +- 3.95
```

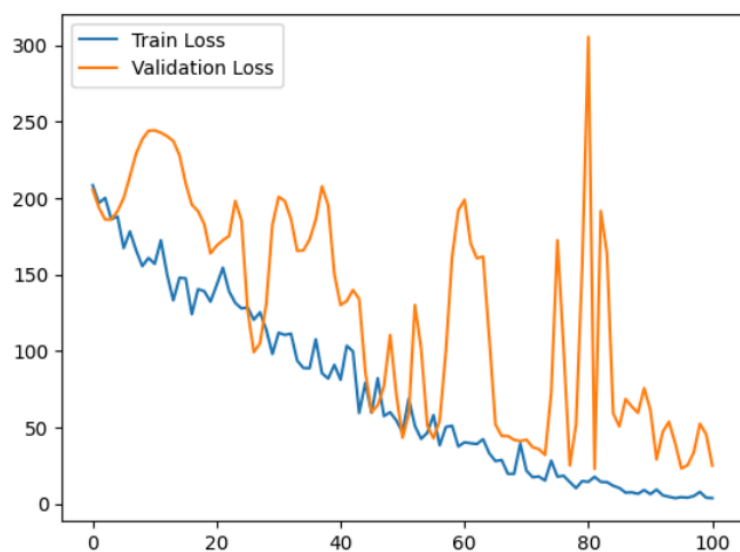
(4) FCN_moreLayers:



```
MSE: 0.0 +- 0.0
RMSE: 0.01 +- 0.0
MAE: 0.01 +- 0.0
Time (sec): 26.33 +- 9.92
```

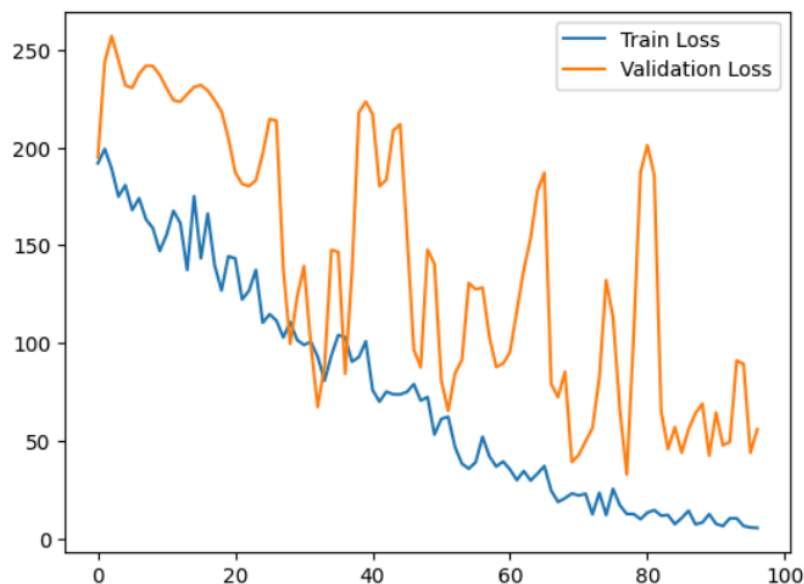
iii) AppliancesEnergy (multivariado):

(1) FCN:



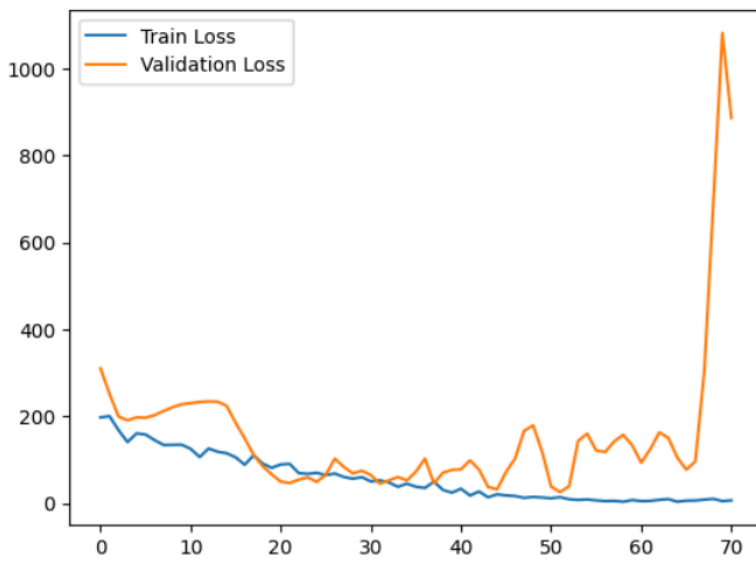
```
MSE: 90.31 +- 48.46
RMSE: 9.1 +- 2.74
MAE: 8.21 +- 3.0
Time (sec): 1.77 +- 1.73
```

(2) FCN_biggerKernels:



```
MSE: 133.41 +- 41.34
RMSE: 11.37 +- 2.06
MAE: 9.42 +- 1.78
Time (sec): 1.97 +- 1.47
```

(3) FCN_moreKernels:



```
MSE: 211.81 +- 132.37
RMSE: 13.85 +- 4.47
MAE: 12.52 +- 4.36
Time (sec): 2.31 +- 1.23
```

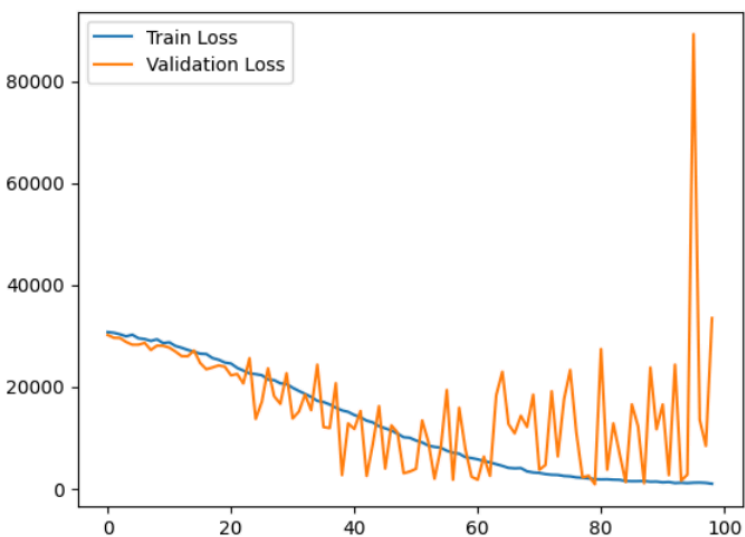
(4) FCN_moreLayers:



```
MSE: 143.15 +- 14.45
RMSE: 11.95 +- 0.6
MAE: 11.43 +- 0.63
Time (sec): 1.78 +- 0.92
```

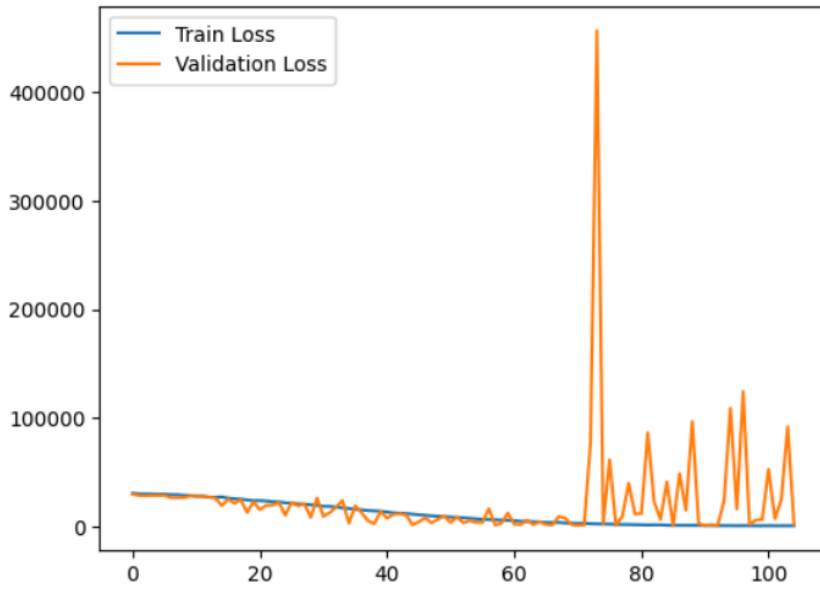
iv) HouseholdPowerConsumption2 (multivariado):

(1) FCN:



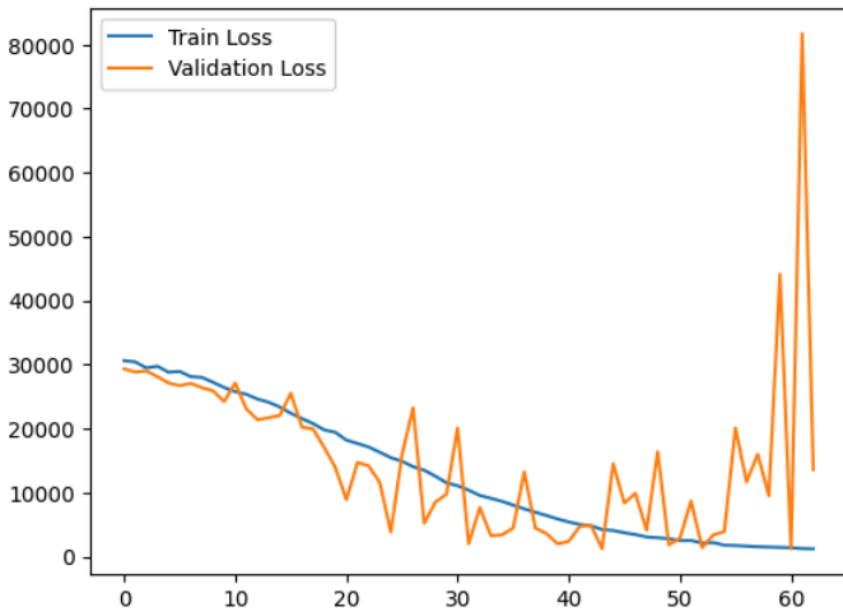
```
MSE: 18557.87 +- 13929.82
RMSE: 127.1 +- 49.03
MAE: 117.79 +- 52.16
Time (sec): 53.54 +- 10.75
```

(2) FCN_biggerKernels:



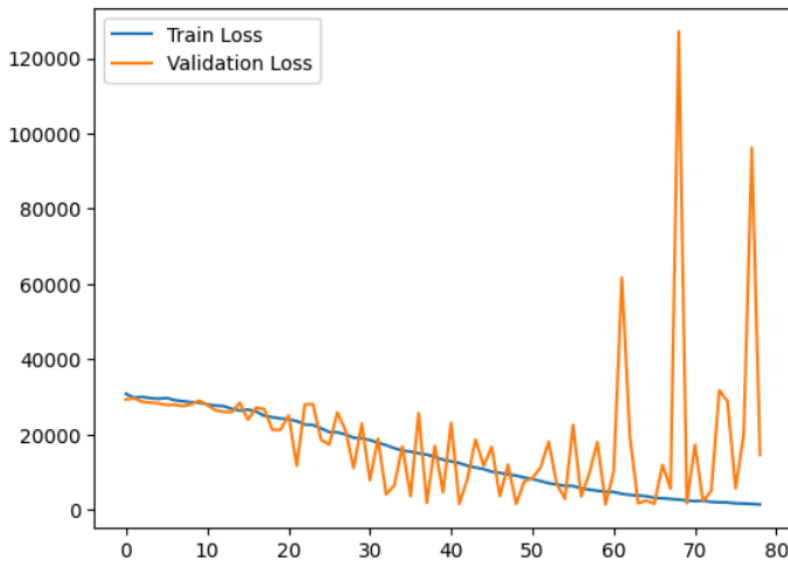
```
MSE: 90847.57 +- 140521.68  
RMSE: 231.98 +- 192.44  
MAE: 224.89 +- 195.04  
Time (sec): 72.74 +- 12.57
```

(3) FCN_moreKernels:



```
MSE: 52267.86 +- 65486.35  
RMSE: 181.16 +- 139.45  
MAE: 172.91 +- 141.89  
Time (sec): 144.74 +- 22.58
```

(4) FCN_moreLayers:



```
MSE: 146606.53 +- 250744.53  
RMSE: 268.08 +- 273.39  
MAE: 261.0 +- 275.6  
Time (sec): 221.11 +- 25.53
```